Derivative-enhanced online estimation from partial observations

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From output derivatives to a convergent online update

The output jet reveals parameters hidden from the raw signal. Its total sensitivity turns that information into a locally convergent update.

Partial signal $y = Rz^\star$ → **Derivative data** $y_m^+ = (\partial_t y, \dots, \partial_t^m y)$ → **Total sensitivity** $\widehat{H}_m = D_\theta \Psi_{m,q} + D_z \Psi_{m,q} \widehat{S}$ → **Local convergence** PE + stability + small gain

Model and output jet

Let Z, Θ and Y be Hilbert spaces, $R \in \mathcal{L}(Z, Y)$, and let the control q be fixed and known:

$$\partial_t z^\star = F_q(t, z^\star, \theta^\star), \qquad y = Rz^\star, \qquad \partial_t \theta^\star = 0.$$

For $m \geq 1$, repeated use of the model defines the positive-order output jet

$$\Psi_{m,q}(t, z, \theta) = (\partial_t Rz, \dots, \partial_t^m Rz).$$

The measured derivative record is denoted by y_m^+.

Sensitivity-aware estimator

The raw-output residual r corrects the state, while the jet residual ρ_m corrects the parameter:

$$r = R\hat{z} - y, \qquad \rho_m = \Psi_{m,q}(t, \hat{z}, \hat{\theta}) - y_m^+.$$

The estimator propagates

$$\begin{aligned}
\partial_t \hat{z} &= F_q(t, \hat{z}, \hat{\theta}) - \mu R^* r, \\
\partial_t \widehat{S} &= [D_z F_q(t, \hat{z}, \hat{\theta}) - \mu R^* R]\widehat{S} + D_\theta F_q(t, \hat{z}, \hat{\theta}), \\
\widehat{H}_m &= D_\theta \Psi_{m,q}(t, \hat{z}, \hat{\theta}) + D_z \Psi_{m,q}(t, \hat{z}, \hat{\theta})\widehat{S}, \\
\partial_t \hat{\theta} &= -\beta \widehat{H}_m^* \rho_m.
\end{aligned}$$

Here R^* and $\widehat{H}_m^*$ are Hilbert adjoints in the chosen fixed metrics, while $\mu, \beta > 0$ are observer and adaptation gains. The internal variable $\widehat{S} \in \mathcal{L}(\Theta, Z)$ is the sensitivity of the output-injected state trajectory to a frozen parameter perturbation. Thus $\widehat{H}_m$ is the complete first-order map

$$\delta\theta \longmapsto \delta\Psi_{m,q}$$

after both the direct parameter effect and the induced state motion are included. Consequently, $-\widehat{H}_m^* \rho_m$ is the local steepest-descent direction for the jet mismatch along the observer trajectory.

Why local exponential convergence holds

Along $(z^\star, \theta^\star)$, let $S^\star$ solve the sensitivity equation with the reference initialisation used in the theorem, and assume it is uniformly bounded; set $H_m^\star = D_\theta \Psi_{m,q} + D_z \Psi_{m,q} S^\star$ there. The proof combines four transparent hypotheses.

1. **The raw-output observer is exponentially stable.** At the true parameter, the evolution generated by
$$A_z^\star(t) = D_z F_q(t, z^\star, \theta^\star) - \mu R^* R$$
satisfies $\|\mathcal{U}_z(t, s)\| \leq C_z e^{-\omega_z (t-s)}$ for $t \geq s$. Thus state perturbations decay when the parameter is frozen at the truth.
2. **The total sensitivity is persistently exciting (PE).** There are $T, \alpha > 0$ such that, for every parameter direction ξ,
$$\int_t^{t+T} \|H_m^\star(s)\xi\|^2 \, ds \geq \alpha \|\xi\|_\Theta^2 \qquad (t \geq 0).$$
Hence no admissible parameter direction remains invisible throughout a full observation window, and the frozen-state parameter equation is exponentially stable.
3. **The nonlinear model is regular near the truth.** F_q and $\Psi_{m,q}$ are locally $C^{1,1}$ with uniformly bounded derivatives, and the augmented estimator is locally well posed with continuation while it stays in this neighbourhood. Its Taylor remainder is quadratic in the errors.
4. **The two stable mechanisms are compatibly coupled.** The jet-induced state self-coupling must first leave a positive observer decay margin. Let $g_{z \leftarrow \theta}$ and $g_{\theta \leftarrow z}$ be the remaining cross-gains from parameter to state and back. The strict condition
$$g_{z \leftarrow \theta}\, g_{\theta \leftarrow z} < 1$$
means that their feedback cannot overcome the two decay margins above.

Theorem. If $\|\hat{z}(0) - z^\star(0)\| + \|\hat{\theta}(0) - \theta^\star\| + \|\widehat{S}(0) - S^\star(0)\|$ is sufficiently small, then the estimator is global and, for some $C_{\mathsf{e}}, \omega > 0$,

$$\|\hat{z}(t) - z^\star(t)\| + \|\hat{\theta}(t) - \theta^\star\| + \|\widehat{S}(t) - S^\star(t)\| \leq C_{\mathsf{e}} e^{-\omega t}\, \mathcal{E}(0),$$

where $\mathcal{E}(0)$ is the same sum of initial errors.

Caveat and outlook

Infinite-dimensional obstruction. With finite-dimensional sensors and norm-continuous $H_m^\star(t)$, the finite-window operator $\xi \mapsto [t \mapsto H_m^\star(t)\xi]$ is compact, hence cannot be bounded below on an infinite-dimensional parameter space. Strong-norm PE therefore requires a finite-dimensional trial space, a weaker parameter norm, or an explicit unresolved-tail bound.

Next steps:

- replace exact derivatives by causal filtered jets with a noise model;
- inject jet innovations into the state observer;
- optimise the MRI control for total-sensitivity PE;
- lift the semi-discrete theorem to the PDE level.

Bloch--Torrey: four quantitative MRI parameters

Let $M_\perp = (M_x, M_y)$ be the measured transverse magnetisation and M_z the hidden longitudinal component. We estimate

$$\theta = (r_2, r_1, \omega, \rho), \qquad T_j = r_j^{-1}, \qquad M_{\mathsf{eq}} = \rho / r_1, \qquad \Delta f = \omega/(2\pi), \qquad y = M_\perp = R(M_\perp, M_z),$$

where C is the transverse readout operator.

What the raw signal sees
The transverse dynamics directly expose r_2 and ω, but do not separate the longitudinal pair r_1 and ρ.

What the second derivative adds
During a known RF pulse $h(t)$, $\delta(\partial_t^2 y)$ contains the longitudinal term

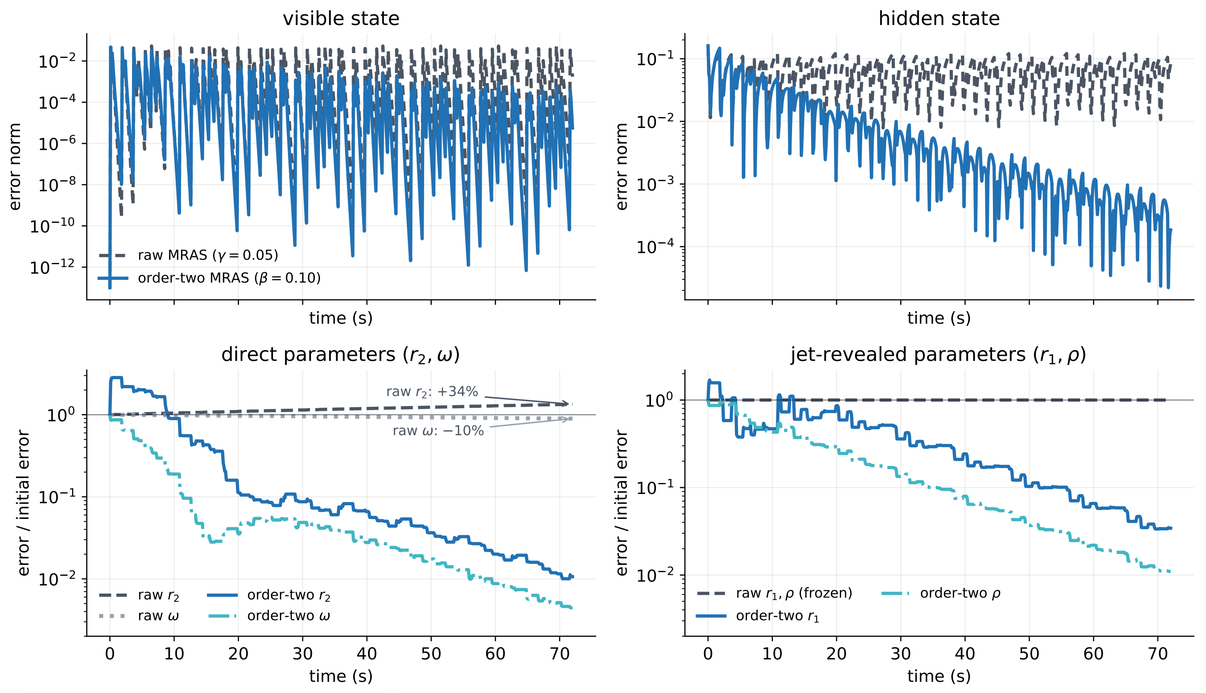
$$R[h(t)(\delta\rho - M_z\, \delta r_1)]\,.$$

Temporal variation of M_z across the sequence separates r_1 from ρ.

Raw updates versus order-two convergence

One-dimensional Neumann Bloch–Torrey model; three known tissue regions, twelve unknown coefficients; standard multi-TI control; identical local initial guess; $\widehat{S}(0) = 0$ for the order-two loop.

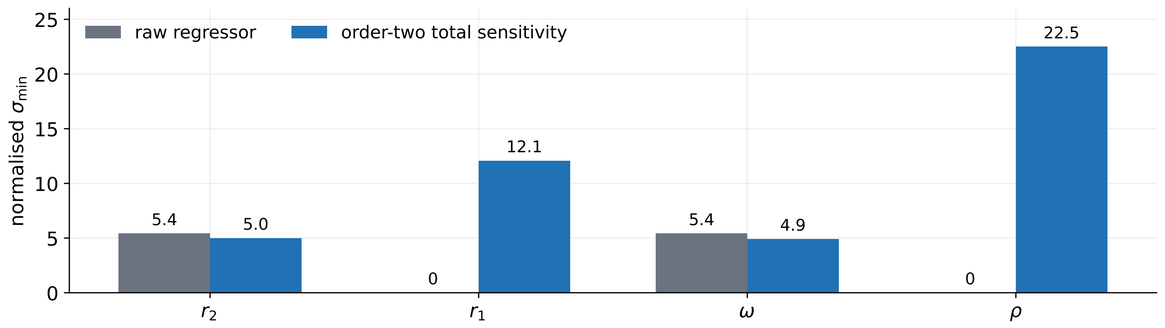
Three-region multi-TI reconstruction from the same local initial guess



Why raw r_2 worsens here. Frozen errors in r_1 and ρ create an amplitude mismatch that raw MRAS misattributes mainly to r_2; its phase-like ω channel is much less contaminated.

Raw MRAS does update r_2 and ω, but not towards joint convergence: their field errors change by $+34\%$ and -10%, while r_1 and ρ are exactly frozen. With the same state observer, the order-two law reduces the final r_2, r_1, ω, ρ errors to $1.1\%, 3.4\%, 0.44\%, 1.1\%$ of their initial values; the hidden-state error falls to 0.11%.

Information gained from the output jet



Bars show $\sigma_{\min}$ of the time-stacked raw regressor and $H_2^\star$, restricted to each field's three regional coefficients over one $36\,\mathrm{s}$ control supercycle. Heights depend on the field and derivative scalings; full twelve-coefficient ranks are $6/12$ (raw) and $12/12$ (order two).

Selected references

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